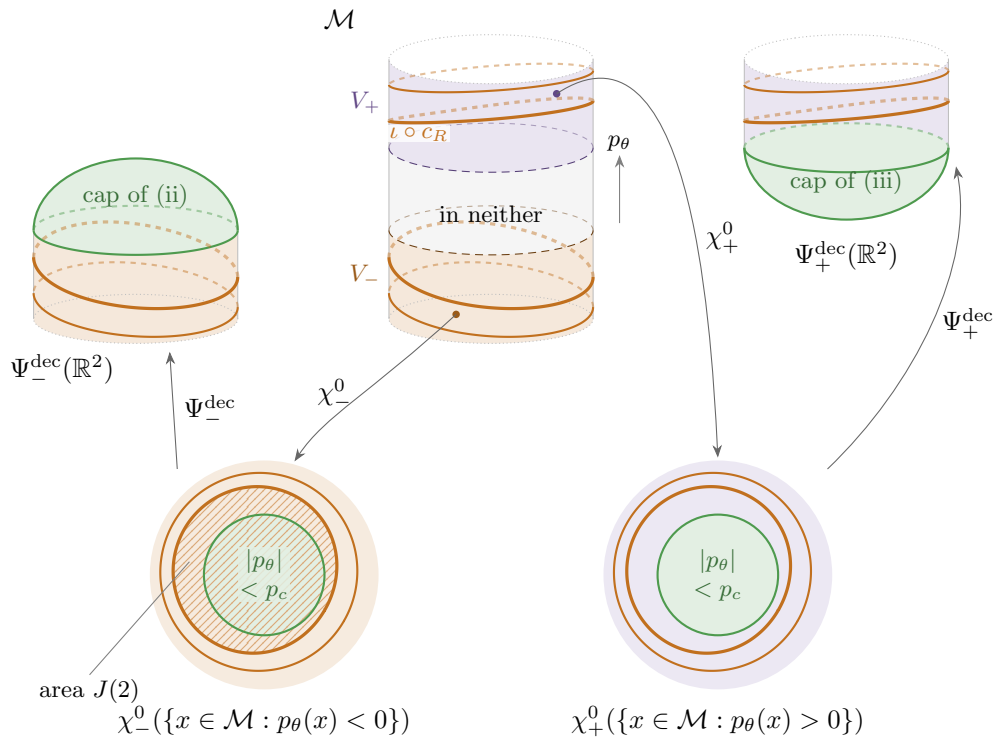

$\mathcal{M}$
V_+
$\iota \circ c_R$
p_θ
in neither
V_-
cap of (ii)
$\Psi_-^{\mathrm{dec}}(\mathbb{R}^2)$
Ψ_-^{dec}
χ_-^0
χ_+^0
cap of (iii)
$\Psi_+^{\mathrm{dec}}(\mathbb{R}^2)$
Ψ_+^{dec}
$|p_\theta| < p_c$
$|p_\theta| < p_c$
area $J(2)$
$\chi_-^0(\{x \in \mathcal{M} : p_\theta(x) < 0\})$
$\chi_+^0(\{x \in \mathcal{M} : p_\theta(x) > 0\})$